Venkata Lakshmi Parimala Pasupuleti

Senior Full-Stack Product Engineer · PropTech · Property Management SaaS · Python · FastAPI · React · TypeScript · PostgreSQL · AWS · OpenAI API · AI Agents · Event-Driven

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PROFESSIONAL SUMMARY

8 years shipping full-stack property management SaaS at Entrata — reservations, pricing calculations, financial settlement, tenant communications, maintenance workflows, document management, and owner reporting at 100K+ unit scale. Owned delivery end-to-end: schema design, backend, frontend, integration, production monitoring, incident response. Base360 is building the same platform for short-term rentals: same systems, same operational problems, different property type. Full-stack (Python + FastAPI + React + TypeScript + PostgreSQL + AWS + OpenAI API + Docker + GitHub Actions) = exact stack. Built Leasel (8-agent LangGraph AI workflow for property operations) and AgentProbe (open-source AI eval infrastructure). M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models and data pipelines; APIs surfacing business insights into application-facing reporting layers.

Magizhs Tech Solution May 2025 – Aug 2025

AI/ML Engineer Intern

- Built agentic automation system (LangGraph + FastAPI) for real estate and commercial survey stakeholders — shipped to real client, iterated weekly on customer feedback. Small team, fast pace, full ownership.

Entrata (PropTech / Real Estate SaaS) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Shipped 20+ full-stack production modules end-to-end** (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints — reservations, pricing calculations, financial settlement, tenant communications, owner reporting, document management, and maintenance workflows for 100K+ housing units. Owned each module from customer problem through schema design, backend, frontend, integration, and production monitoring.
- **Owned the core pricing and eligibility pipeline for 7 years** — annual data pipeline computing rent calculations, HAP payments, income-limit eligibility, and tax-credit thresholds. Same system as Base360's dynamic pricing and financial reporting layer: rules-based computation, regulatory compliance, owner statements. Zero compliance violations.
- **Built event-driven automation system (RabbitMQ)** for triggering account state updates across thousands of records — retry logic, idempotency guarantees, owned through production incidents. Same pattern as "booking received → trigger cleaning schedule → send check-in instructions."
- **Owned multi-tenant PostgreSQL schemas** across pricing, financial settlement, document management, and tenant communication workflows — 200+ SQL queries optimized (30% gain); schema design supporting data isolation across 100K+ units and multiple property management organizations.
- Microservices redesign (Docker) — 50% scalability improvement, 40% deployment time reduction; mentored 5+ engineers across cross-functional product, engineering, and compliance teams.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built property management modules (Python, Java, PostgreSQL) across reservations, document storage, and compliance workflows; led AWS S3 migration; redesigned document storage architecture for hundreds of thousands of records.

PROJECTS

Leasel: AI-Powered Property Operations Product PROPTECH AI FULL PRODUCT

Python · LangGraph (8-agent) · FastAPI · OpenAI API · RAG · OCR · Streamlit | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 3-surface AI platform for property operations: (1) Document Intelligence — OCR ingestion, section-aware chunking, hybrid search, citation grounding; (2) Support Ticket Triage — 8-agent LangGraph workflow routes tickets to specialized agents (configuration check, historical incident lookup, document analysis), produces evidence-backed conclusions, escalates to human when confidence is low; (3) Tenant Self-Service — policy questions answered directly; eligibility always routes to staff. Same AI-powered communication and triage pattern as guest communication + maintenance request automation.
- Eval: 93.6% precision, 87.9% F1 on 1,240 lease PDFs. Core rule: LLM reasons; human or deterministic logic makes every final regulated call.

AgentProbe: AI Evaluation Infrastructure OPEN SOURCE CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend, 5-metric eval engine, CI regression harness with golden datasets, model routing across OpenAI/Anthropic/Groq/Ollama. Measures AI-powered workflows before they reach production users — the harness that confirms AI features work before operators see them.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), JavaScript, TypeScript, Java, SQL, R, Bash — FastAPI, React.js, Django, Spring Boot, REST APIs

AI & Automation: LangChain, LangGraph, OpenAI API, Anthropic API, RAG Pipelines, Agentic Workflows, Tool Calling, HITL, Embeddings, ChromaDB, Prompt Engineering, LLM Evaluation, Model Routing, AI Agents, Workflow Automation

Backend & Data: PostgreSQL, Supabase, Redis, RabbitMQ (event-driven), Apache Kafka, Celery, Temporal, Microservices, Multi-Tenant Architecture, Distributed Systems, Idempotency, Payments Integration, Financial Settlement, ETL, Data Pipelines, Product Analytics

Cloud & Infra: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry, Serverless Architecture